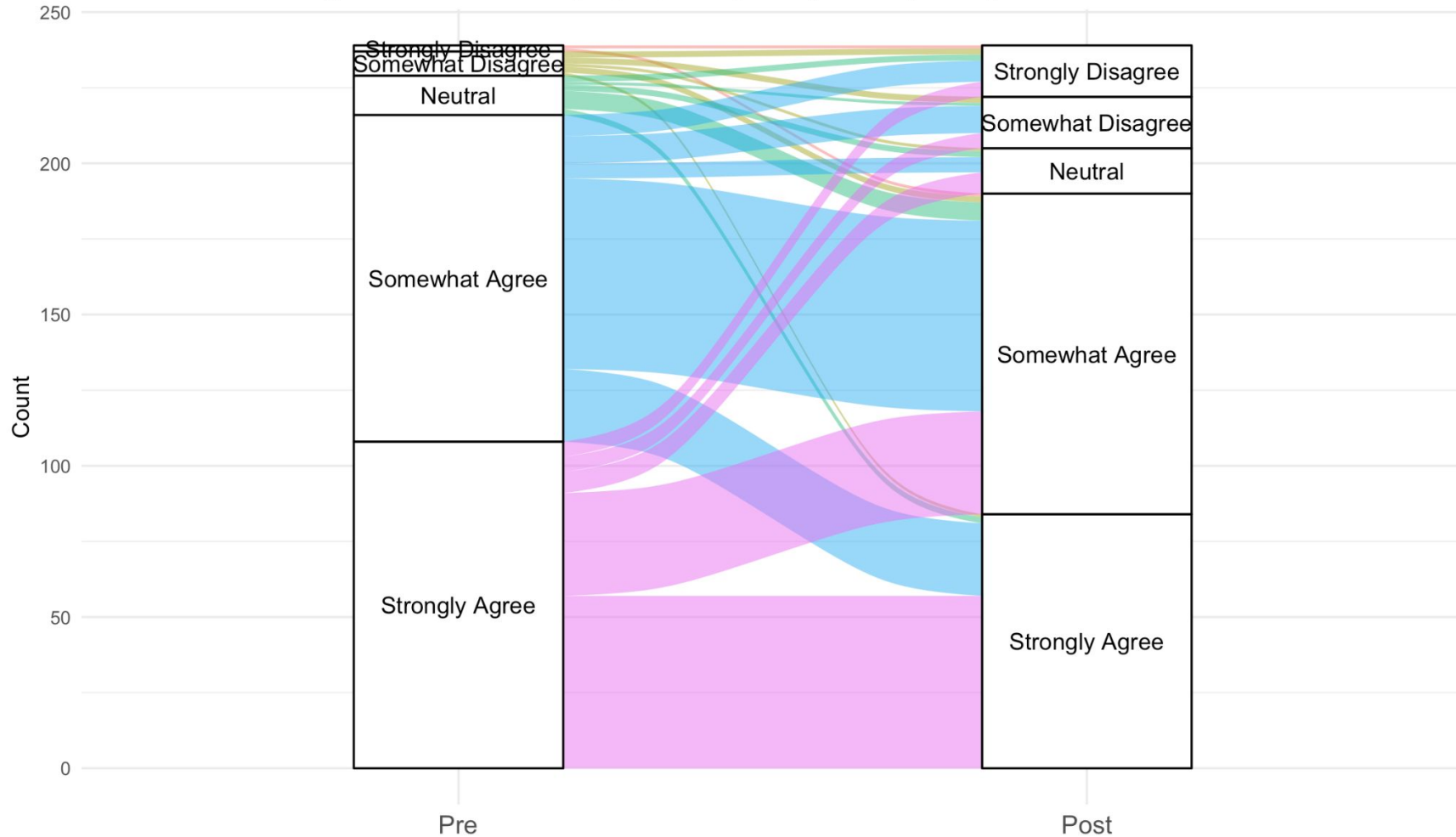


Course Announcements

Due Sunday:

- D3 - tests provided are too strict; we will manually grade
- Q4
- A2 - use `choco_fruty` (not `i`) throughout (not `fruity_choco`)
- [Previous Project Review](#)

Pre & Post: When working on a data science project, data privacy is the primary ethical concern.



Data Science Questions

Shannon E. Ellis, Ph.D
UC San Diego



Department of Cognitive Science
sellis@ucsd.edu

Formulating Data Science Questions

When you and your group sit down to figure out what you're going to do for your final project in this class, you'll have to formulate a strong question - one that is specific, can be answered with data, and makes clear what exactly is being measured.

Nature of a data scientist

- data-driven
- care about answers
- care about whether the results make sense, because they care about what the answers mean.
- are comfortable with the idea that data have errors
- are comfortable with the idea that there's more than one way to analyze the same data
- know nothing is ever completely true or false in science
- know that you can still learn something and make decision despite the uncertainty
- care about communicating these subtleties *and* the results

Nature of a (GREAT) data scientist

- Conscientious - works using proven and understood methods; triple checks things
- Open & Creative - open to new methods, creative at finding solutions (just checks them thoroughly)
- Methodical
- Big Picture in view - can take a step back from the details to see/question the big picture

The Data Science Process

Ask an interesting question.

What is the scientific goal?
What would you do if you had all the data?
What do you want to **predict** or **estimate**?

Get the data.

How were the data sampled?
Which data are **relevant**?
Are there **privacy** issues?

Explore the data.

Plot the data.
Are there **anomalies**?
Are there **patterns**?

Model the data.

Build a model.
Fit the model.
Validate the model.

Communicate and visualize the results.

What did we learn?
Do the results make sense?
Can we tell a story?

Joe Blitzstein and Hanspeter Pfister, created for the Harvard data science course <http://www.cs109.org/>.

Asking Good Questions is Key!

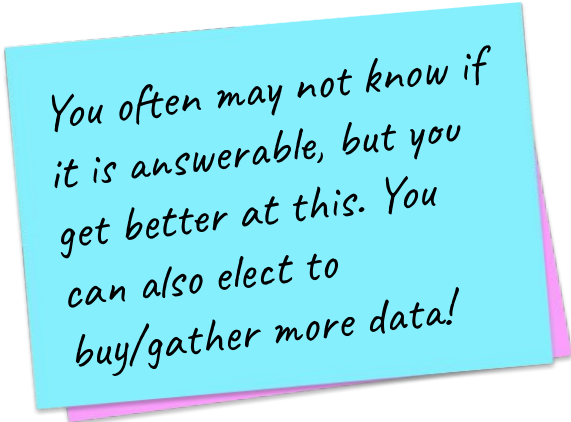
If I had an hour to solve a problem and my life depended on it, I would use the first 55 minutes determining the proper question to ask, for once I know the proper question, I could solve the problem in less than five minutes.

—Einstein



Data Science questions should...

- Be answerable with (available/attainable) data
- Unambiguous in meaning
- Specify what's being measured
- Have relevance to someone or something
- Big enough to be interesting; small enough to be accomplishable



You often may not know if it is answerable, but you get better at this. You can also elect to buy/gather more data!

Specifying what you're going to measure is important:

Examples of poor questions that leave wiggle room for useless answers:

- What can my data tell me about my business?
- How can I increase my profits?

Examples of good questions where the answer is impossible to avoid:

- How many Cybertrucks will Tesla sell in San Diego during the third quarter?
- What is the relationship between how many students are offered admissions at UCSD and how many enroll?

**Working toward a strong data
science question...**

Nailing down the right question: **politics**

Too-vague: What impacts American politics?

... Does pop culture have an impact on American politics?

... Do American TV shows have an *impact* on American politics?

... Does South Park *affect* American politics?

... Is there a *relationship between* words in South Park episodes and American politics?

... Is there a relationship between the sentiment of political words in South Park and American politics?

Better: What is the relationship between the sentiment of political words in South Park and America's presidential approval rating?

Nailing down the right question: education

Too-vague: How has COVID-19 impacted students?

Improving: How has COVID-19 impacted university students' education?

... Do students' grades and how they rate their classes differ pre- and during remote learning, due to COVID-19?

... At UCSD, is there a difference between students' grades and how they rate their classes before COVID-19 and during remote learning, due to COVID-19?

Better: At UCSD, comparing remote learning during COVID-19 to the two years pre-COVID, what was the effect on students' grades and how students rate their courses?

Data Science Question

Here is a too-vague question:

What racial disparities exist in
policing?

Let's refine it...Which of the
following would be the best
possible data science question?



Think-Pair-Share: San Diego CoStS

Too vague question: Why is it so expensive to live in San Diego?

9AM:

2PM:

What happens next?

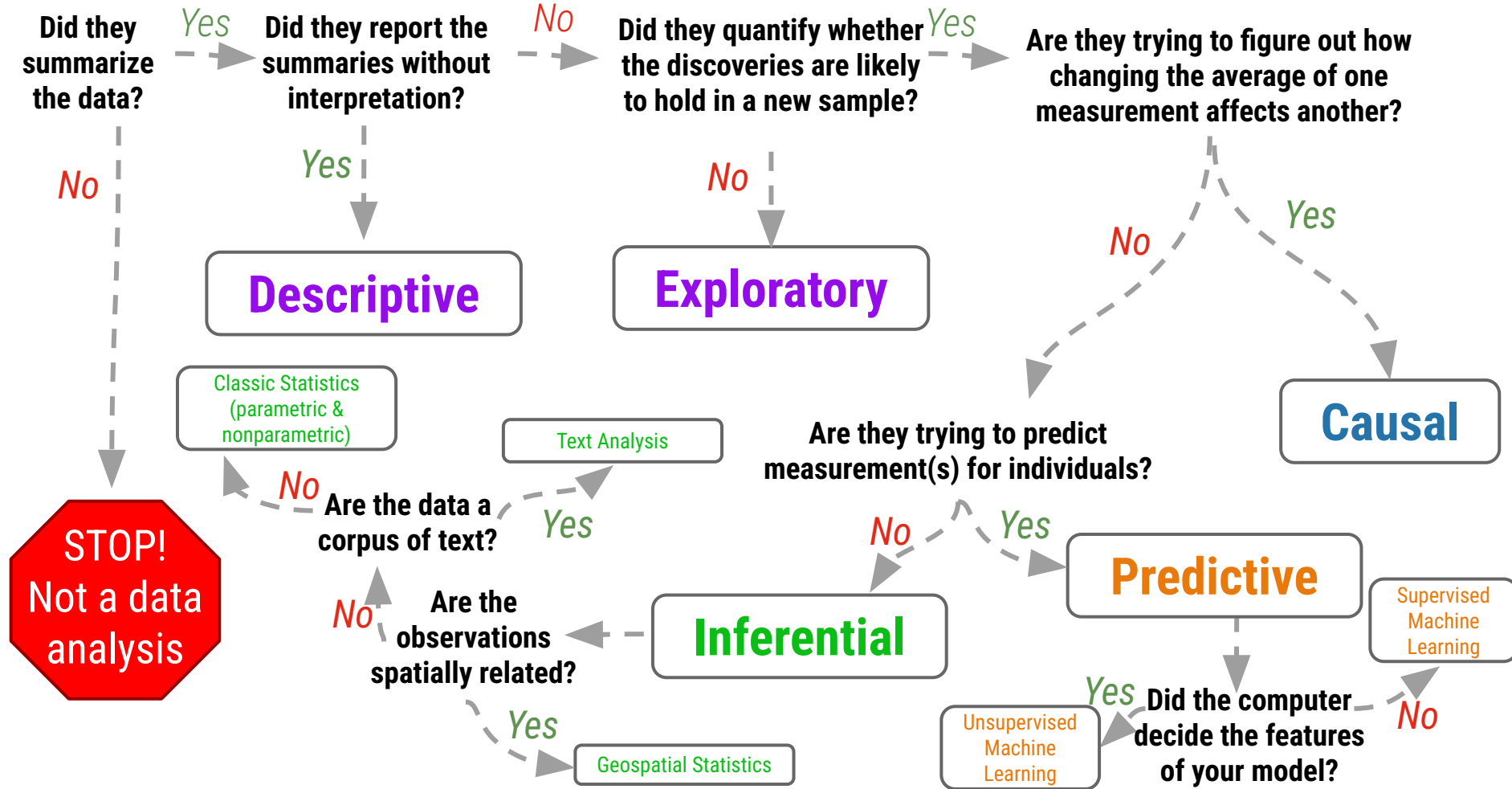
After the question is defined, should it become a project?

- What are the constraints?
- What are the resources available?
- ARE THE NECESSARY DATA GETTABLE? - will need this for first checkpoint
- Costs? Benefits?
- Potential risks? Rewards? (Ethical)
- Can we define a metric to determine success of the project?

*“Data science is the process of formulating a quantitative question that can be answered with data, collecting and cleaning the data, **analyzing the data**, and communicating the answer to the question to a relevant audience.”*

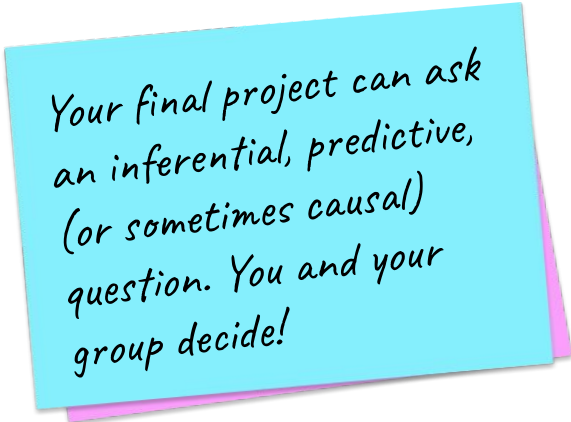


To do this, you have to look at, describe, and explore the data



Summary: Analytical Approaches

1. **Exploratory** Data Analysis is the first step
2. **Inference** establishes relationships
 - a. Classic Statistics
 - b. Geospatial Analysis
 - c. Text Analysis
3. Machine Learning is for **prediction**
 - a. Supervised
 - b. Unsupervised
4. Experiments best way to establish **causality**



Your final project can ask an inferential, predictive, (or sometimes causal) question. You and your group decide!

General question: What impacts politics in America?

Data Science question: Is there a relationship between the sentiment of political words in South Park and America's presidential approval rating?

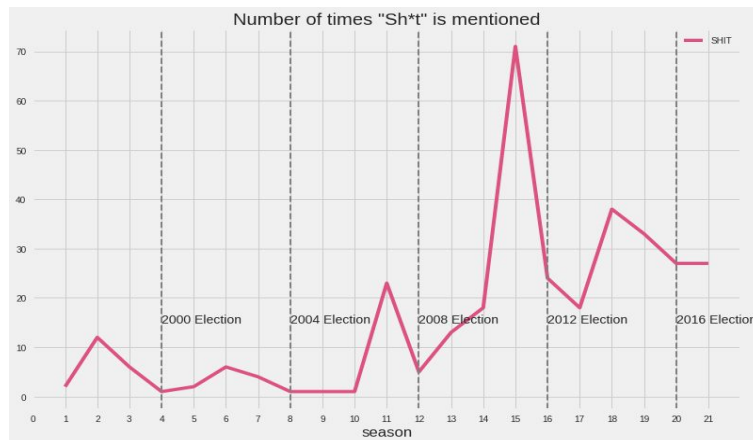
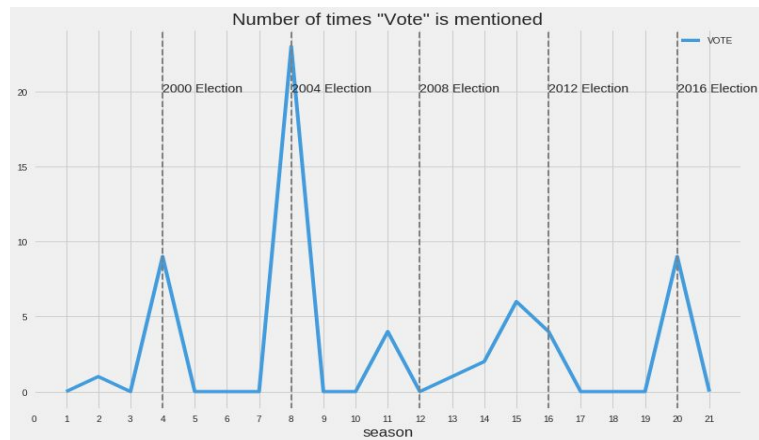
Descriptive

Exploratory

Inferential

Text Analysis

Classic Statistics
(parametric &
nonparametric)



General question: How has COVID-19 impacted students?

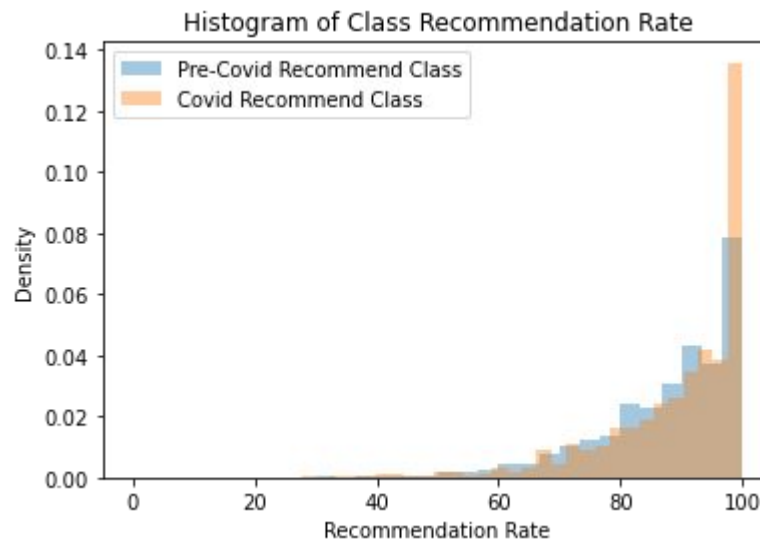
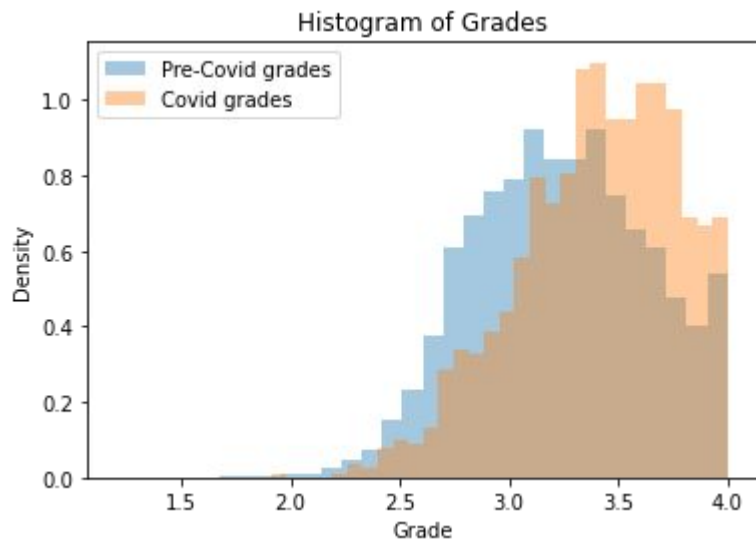
Data Science question: At UCSD, comparing remote learning during COVID-19 to the two years pre-COVID, what was the effect on students' grades and how students rate their courses?

Descriptive

Exploratory

Inferential

Classic Statistics
(parametric &
nonparametric)



General question: Why isn't police response time always the same?

Data Science question: Where should police cars be stationed, accounting for crime levels and time of day, to make police response times equitable throughout San Diego?

Descriptive

Exploratory

Predictive

Inferential

Geospatial Analysis

